NLP Project Proposal

Team ID: 13

1) Devadatta Mahesh Pokharanakar- 142502012
2) Chilaka Sri Krishna Sai - 142502010
3) Vaddi Govardhan - 142502032

Title: Analysis of Indian Parliamentary Proceedings Transcripts Using NLP

Motivation:

Parliamentary proceedings are mostly released as long PDF documents. These are hard to read quickly. Important information such as who spoke, on which topic, and what policies or numbers were mentioned is buried inside many pages of text.

There is no public tool that performs all these analysis steps together in one place. Recent government initiatives such as **Sansad Bhashini** show that automated processing of parliamentary data is becoming important. This motivates us to work in this area.

NLP Tasks / Applications

We plan to build a system that processes English parliamentary session documents such as:

- Budget speeches
- Bill introductions
- Question–Answer sessions
- Debates
- Ministerial statements

The system will automatically:

- Identify the speaker, role, party, and ministry
- Detect topics discussed
- Extract named entities such as people, places, schemes, ministries, bills, and money values
- Detect factual references and policy mentions
- Perform sentiment or stance analysis
- Generate topic-wise and speaker-wise summaries
- Compute time-based statistics
- Produce simple visualizations

The goal is to convert long unstructured documents into structured data.
The system will work on one document at a time.

Approach:

We model the problem as a document-processing pipeline.

1. Pre-processing: Convert PDFs to text, remove headers and footers, normalize spacing, detect timestamps, and segment the text into speaker turns. Rule-based methods will be used as initial baselines.
2. Speaker Identification: We will use pattern-based rules and sequence labeling models such as BiLSTM-CRF and transformer-based token classifiers.
3. Named Entity Recognition: We will detect people, parties, ministries, schemes, bills, locations, organizations, money values, and dates. A general-purpose English NER system will be used as a baseline, and domain-adapted transformer models will be evaluated.
4. Topic Segmentation and Modeling: Long speeches will be divided into policy sections such as agriculture or taxation. We will compare classical methods like TextTiling and LDA with embedding-based clustering approaches.
5. Sentiment / Stance Detection: Statements will be classified into categories such as *Appreciate, Neutral, Call for Action, Blame,* and *Issue*. We will use TF–IDF with logistic regression as a baseline and transformer-based classifiers for improved performance.
6. Summarization: Topic-wise and speaker-wise summaries will be generated. Extractive methods such as TextRank will serve as baselines, while neural sequence-to-sequence models will be explored for abstractive summarization.
7. Output: The final system will produce structured JSON files, tables, timelines, and visual analytics.

Data:

We will use the **Parliament Digital Library (PDL)** as our primary source .

We will collect Lok Sabha and Rajya Sabha proceedings, including Budget sessions, debates, and Question Hour transcripts.

Although pre-trained models exist, parliamentary language is domain-specific. We will therefore manually annotate a small subset of documents to label speakers, topics, named entities, and procedural text if required. This annotated data will be used for fine-tuning and evaluation.

References:

[1] Rohit, Sakala & Singh, Navjyoti. (2018). Analysis of Speeches in Indian Parliamentary Debates. 10.48550/arXiv.1808.06834.

[2] Wijeratne, Yudhanjaya & de Silva, Nisansa & Shanmugarasa, Yashothara. (2019). Natural Language Processing for Government: Problems and Potential. Link

[3]Katre, Paritosh. (2019). NLP Based Text Analytics and Visualization of Political Speeches. International Journal of Recent Technology and Engineering. 8. 8574-8579. 10.35940/ijrte.C6503.098319.